Welcome to the course

From analytics to action

The course at a glance

- Data analytics in real-world organizations
- You bring data science skills
- We bring social science perspectives with special attention to data-driven management and organizational context and change
- Case organizations
  - Health care
  - Tourism
- 13 weeks; project-based in groups, oral exam (poster presentation and discussion)
- Lectures, guest lectures, exercises, group work, reading seminars
- Physical presence is required

The team - lecturers

- Tanja Schneider, associate professor
- Research area: Values in technology, social struggle over value, sociology of markets
- Course coordinator and lecturer

- Anders Koed Madsen, professor
- Research area: Digital methods innovation, soft city sensing, participatory data analytics, Lecturer

- Guest lecturers: Brit Ross Winthereik, professor & Sophie Mützel, professor

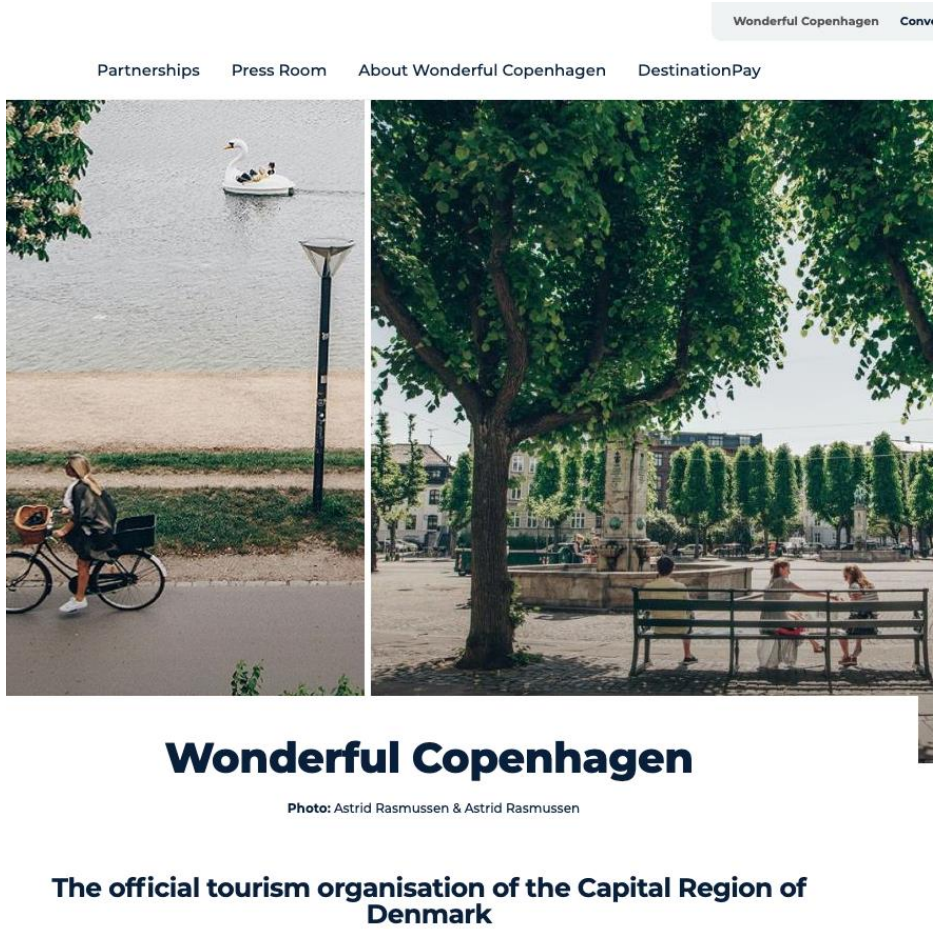
The team – teaching assistants

- **Isabela Maranca:** PhD student, human-decision making, Intelligent Transportation Systems, Transportation Science

- **Johanne Hvidberg Conradsen:** PhD student, human decision making, Intelligent Transportation Systems, Transportation Science

- **Rabbi Rahman**: Part-time lecturer AAU in Techno-Anthropology programme, digital transformation, socio-technical networks

The cases/organisations



Tell us a bit more about your background:

Join at:
vevox.app

ID:
166-237-272

Course objective and learning goals

Goal: The general purpose of the course is to qualify the student to **understand and handle problems and opportunities in data analysis projects from a management perspective.**

Learning goals:

- analyze challenges and opportunities in a specific data analysis project;
- identify the project's stakeholders;
- define the project's commercial value;
- compare and assess potential pitfalls and barriers for the project;
- develop a realistic and well-thought-out plan for the implementation of the project in a concrete company;
- visualize the project's prerequisites, processes and results;
- discuss the project's possible consequences for the market and organisation;
- design a strategy for the involvement of actors who are relevant to the implementation and impact of the project;
- prepare a presentation of the project's opportunities and challenges to senior management;
- communicate the project's results and relevance to the company's business in clear and well-argued language.

Organisational situation

- Build on information provided by your chosen case company
- Conduct general internet research on industry/sector
- Draw on lectures and guest lectures, specifically use analytical concepts introduced in these lectures to think through organisational situation

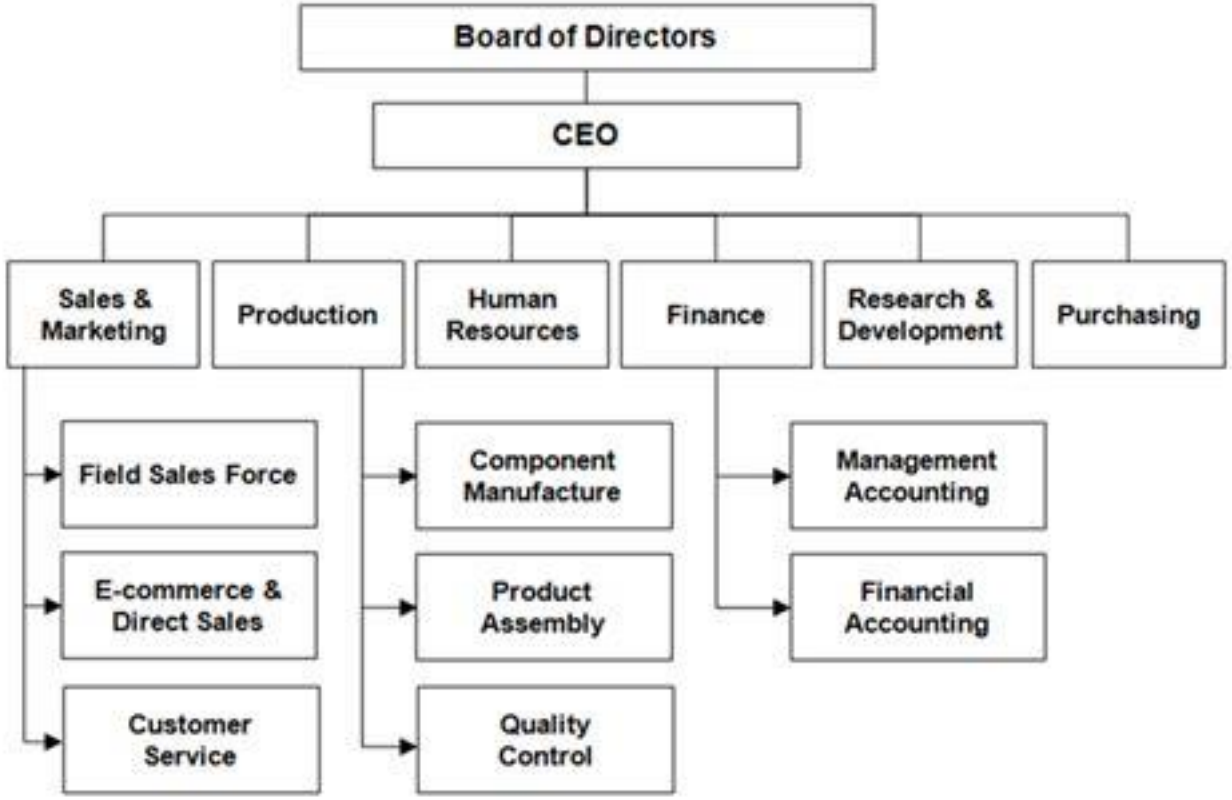
Data analytics

- Build on your pre-existing data analysis skills
- Use data sets provided by case companies
- Employ exploratory data analysis and visualisation as taught in lectures by Anders Koed Madsen

Communication

- Present case opportunities and challenges to senior management
- Clearly communicate results and relevance

Taking the organization into account



Not just information that must be pieced together, but expectations that must be balanced

- With any data set comes sets of values, infrastructures, and power relations
- Hopes and dreams about what is possible with data

Numbers rarely speak for themselves

Cultural bias towards numbers as requiring no proof or explanation; statistics are rarely self-evident.

"In the self-evident method [of inquiry] the presuppositions of our own culture's routine practices are not regarded as in need of explanation" (Shapin and Schaffer, 1985, p. 5)

Baseline for this course: We need to understand not only the technology and the methodology, but also the (cultural) assumptions to provide a good translation from analytics to action

What is culture?

- A system of meaning that can be interpreted (C. Geertz)

- Organizations have culture
  – How they see things
  – How they do things
  – The value they attach to things

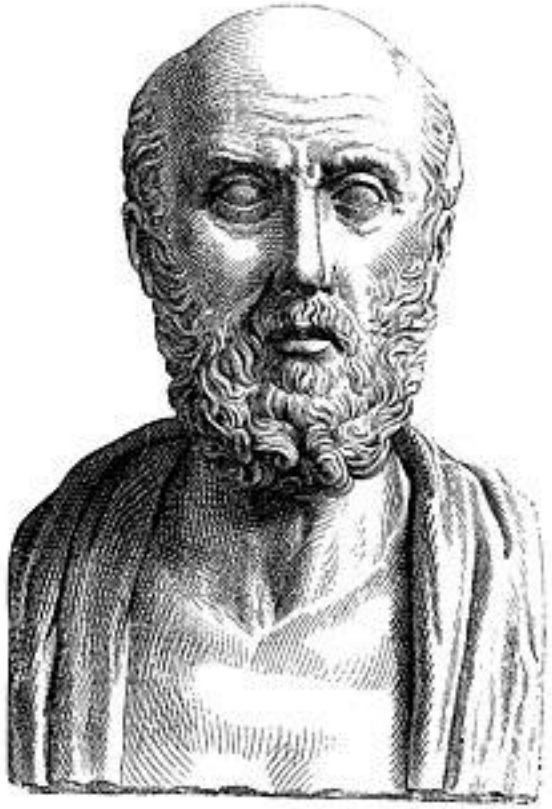
What's the meaning of the boy's wink?

Winking boys

Balinese cock fights

We live in a technological culture

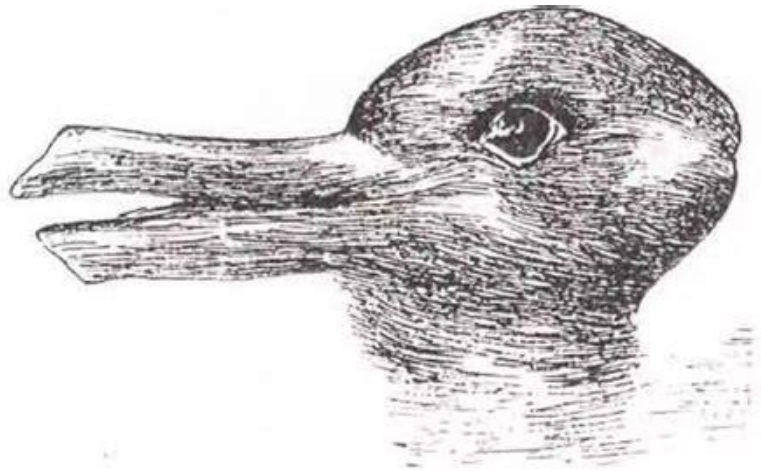
- Digital technologies pervade all aspects of daily lives, work and government
- Technology considered to equal progress
- Data analysts play a crucial role in digitalized societies



Translation is needed

How to translate faithfully when everyone is pre-disposed

- There is no place from where you see the world ‘objectively’
- As data analysts we also have particular vantage points - preferences are hard-coded into our bodies
- Important to understand that different perspectives co-exist when you do data-driven management or advise towards action



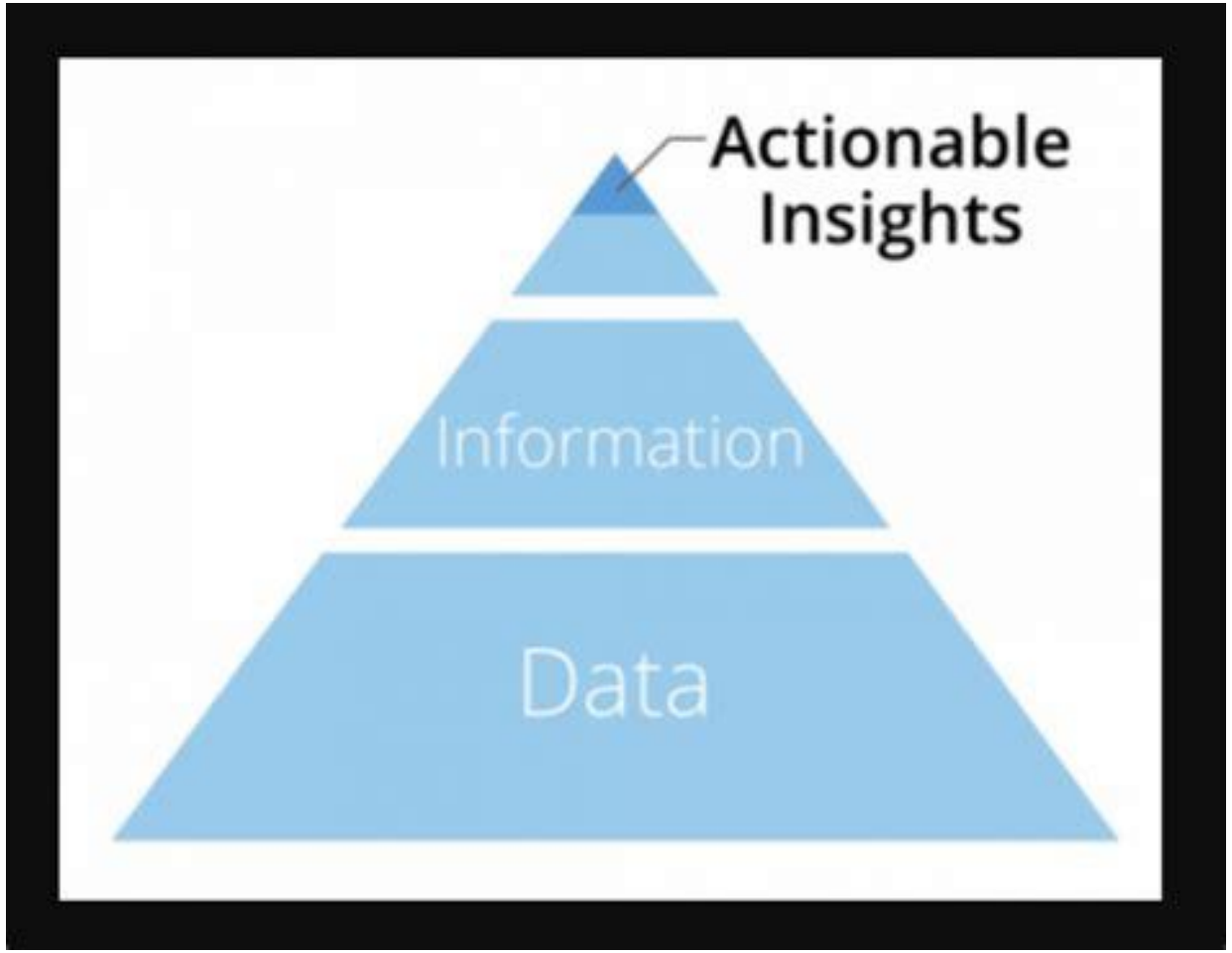
Co-existing perspectives

- Pain by walking
- Thickened vein
- Traits on an x-ray

- A question of coordination

Mol, A. (2002). *The body multiple: Ontology in medical practice*. Duke University Press.

Making data actionable – the standard view



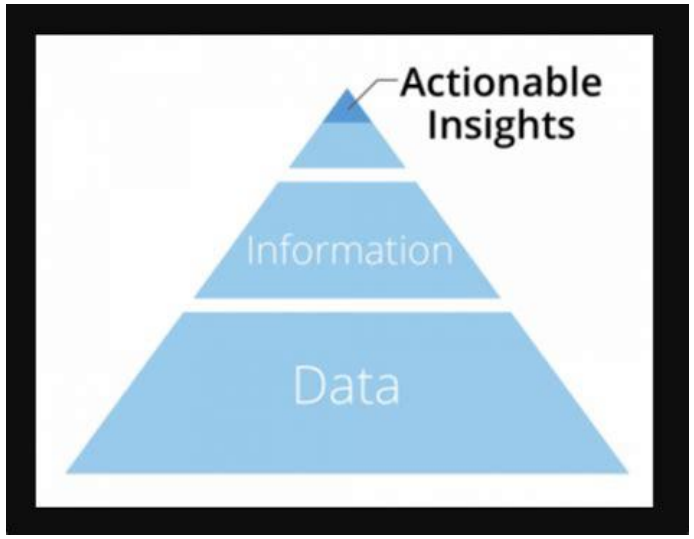
The organisation

Making data actionable – the broader context

Societal trends

Stakeholder x, y, z

The environment



Laws and regulation

Markets

Innovation/Emerging markets

The organization

Mega trends

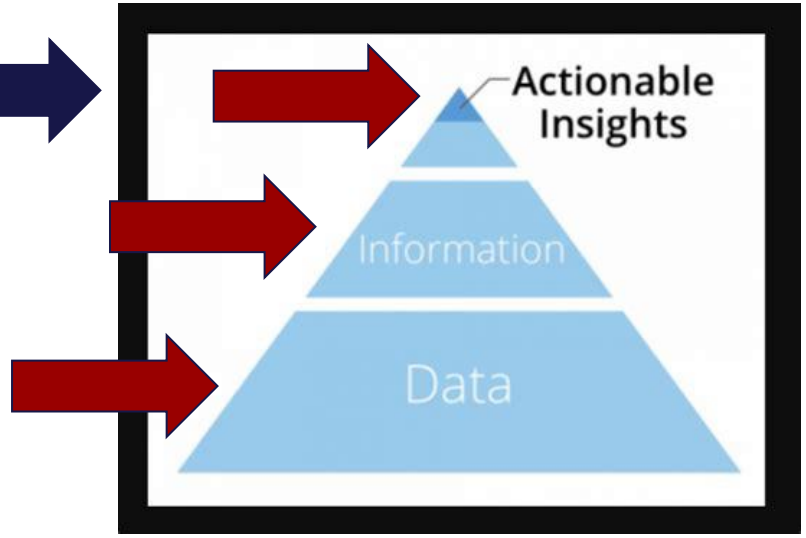
Board of Directors/Governors

Ethics of data analytics

Societal trends

Stakeholder x, y, z

The environment



Laws and regulation

Markets

Innovation/Emerging markets

The organization

Mega trends

Board of Directors/Governors

Data analytics as ethics

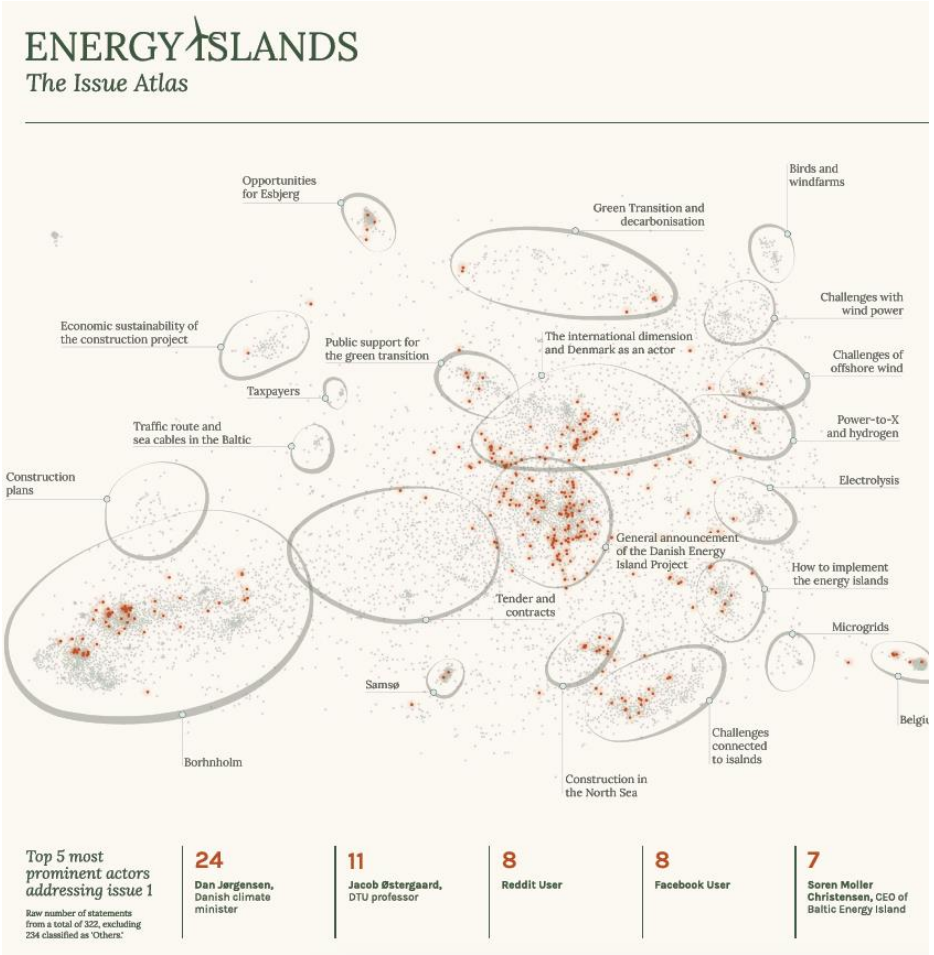
Collecting:
Flawed/biased data

Processing:
Unlawful treatment of data

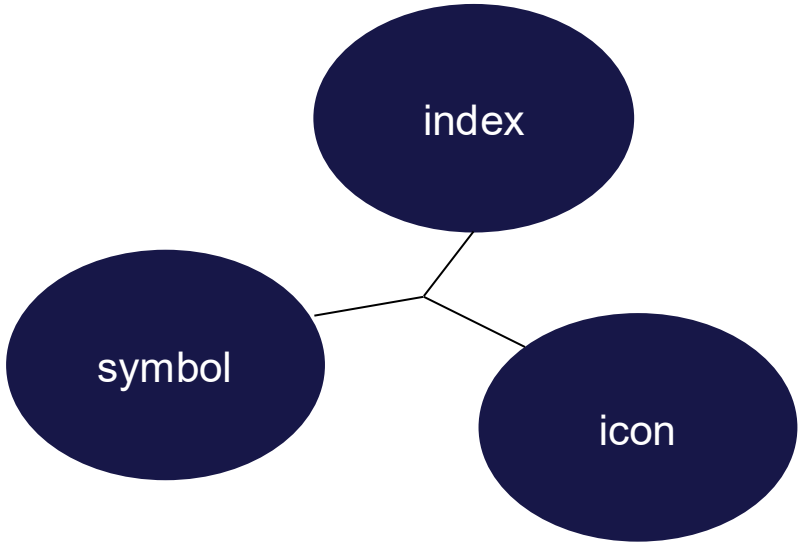
Presenting:
Disagreement as to what counts as valuable

Attending to ethics in your projects

- What to highlight and why
- Translate technical knowledge and turn into communicable insights
- Make transparent how decisions were made in an engaging and transparent manner without ‘overburdening’ the story
- Reflect on your own concerns – they may be valid and a deserve a place in the story



Data-driven management – what's in a number?



A data point as a semiotic structure (symbol, index, icon)

When you learn this, you will understand that data have a world inside them (data is never 'raw'), and can contribute to change if it is 'cooked with care' (*Raw Data" Is an Oxymoron* edited by Lisa Gitelman. The MIT Press)

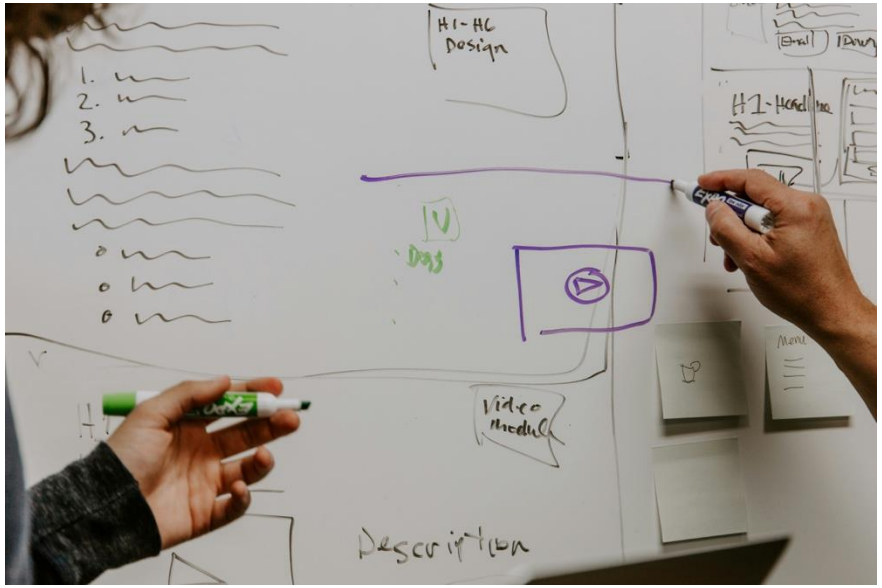
‘Numbers that speak for themselves’ versus ‘Stories that resonate’

- Data analytics reveal (parts of) the organisation and its surroundings to management.
- Data analysts must learn to tell stories to aptly illustrate a situation and equip management to make the best possible use of data.
- Stories and good communication skills are *invaluable* because ‘the best’ (what is most valuable at a given time and place) varies.
- If data analysts are aware of their role as ethical storytellers, it makes for actionable data that resonate with organisational strategies.

Setting expectations

- You are data scientists and know the basics
- You still need to learn how to fit data projects into an organization, what the contingencies might be
  - When you want to make your data relevant YOU need to change ‘at home’, add to your own practice, develop your expertise within organizational change

- We are the social data scientists (sociology and organization studies)
- The course offers new perspectives on data projects and an elaborate exercise that will equip you to deliver back to the case companies



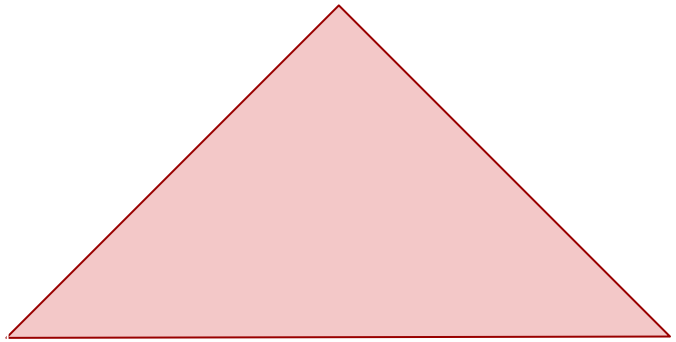
Social data science

- Informed by sociology and organization studies
- Inter-disciplinary
- See technical issues as consisting of both social and technical elements (as sociotechnical)
- Has the capacity to translate across technical and organizational issues

#Technologyforpeople

The ‘data-situation’ in many contemporary organizations

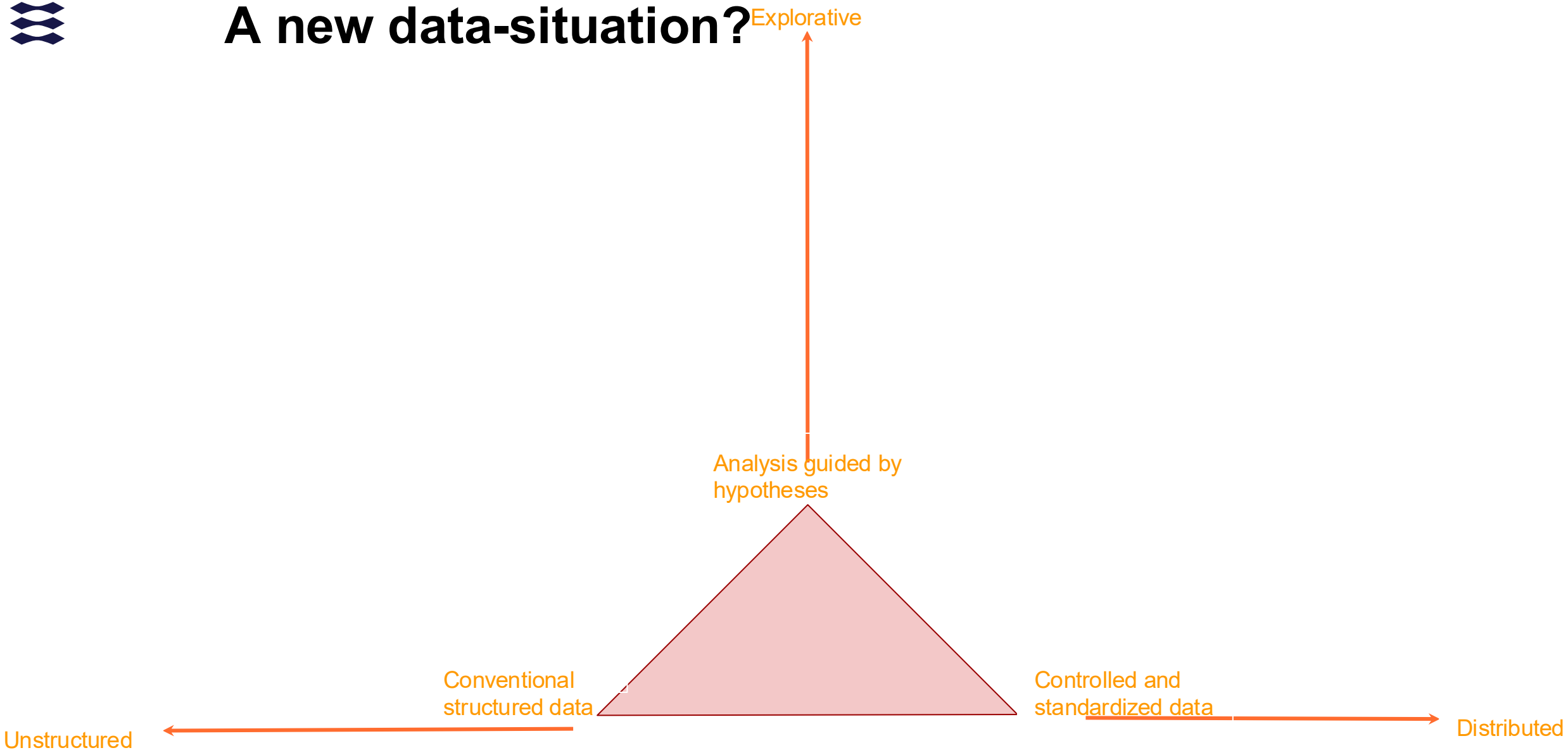
Analysis guided
by hypotheses



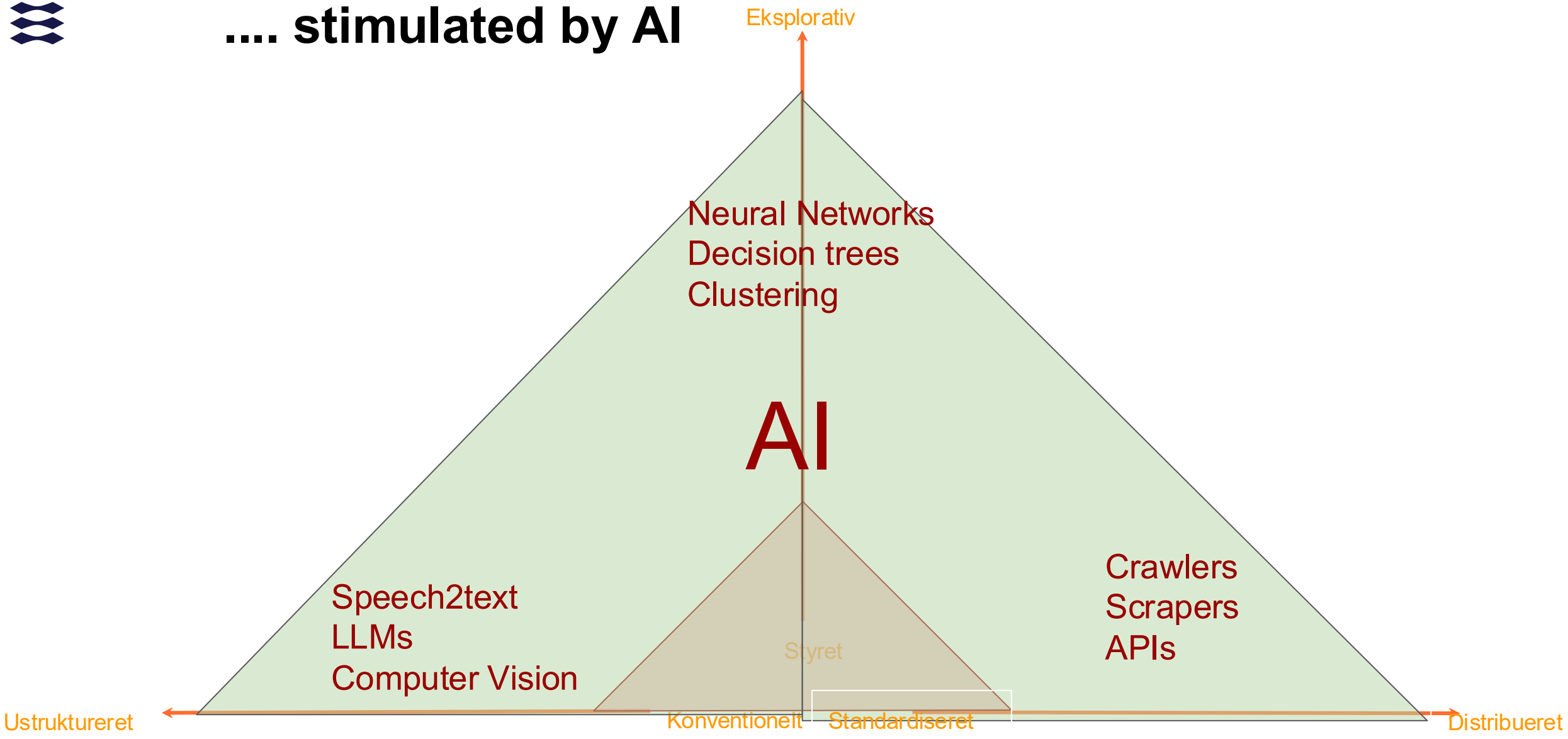
Conventional
structured data

Controlled and
standardized data

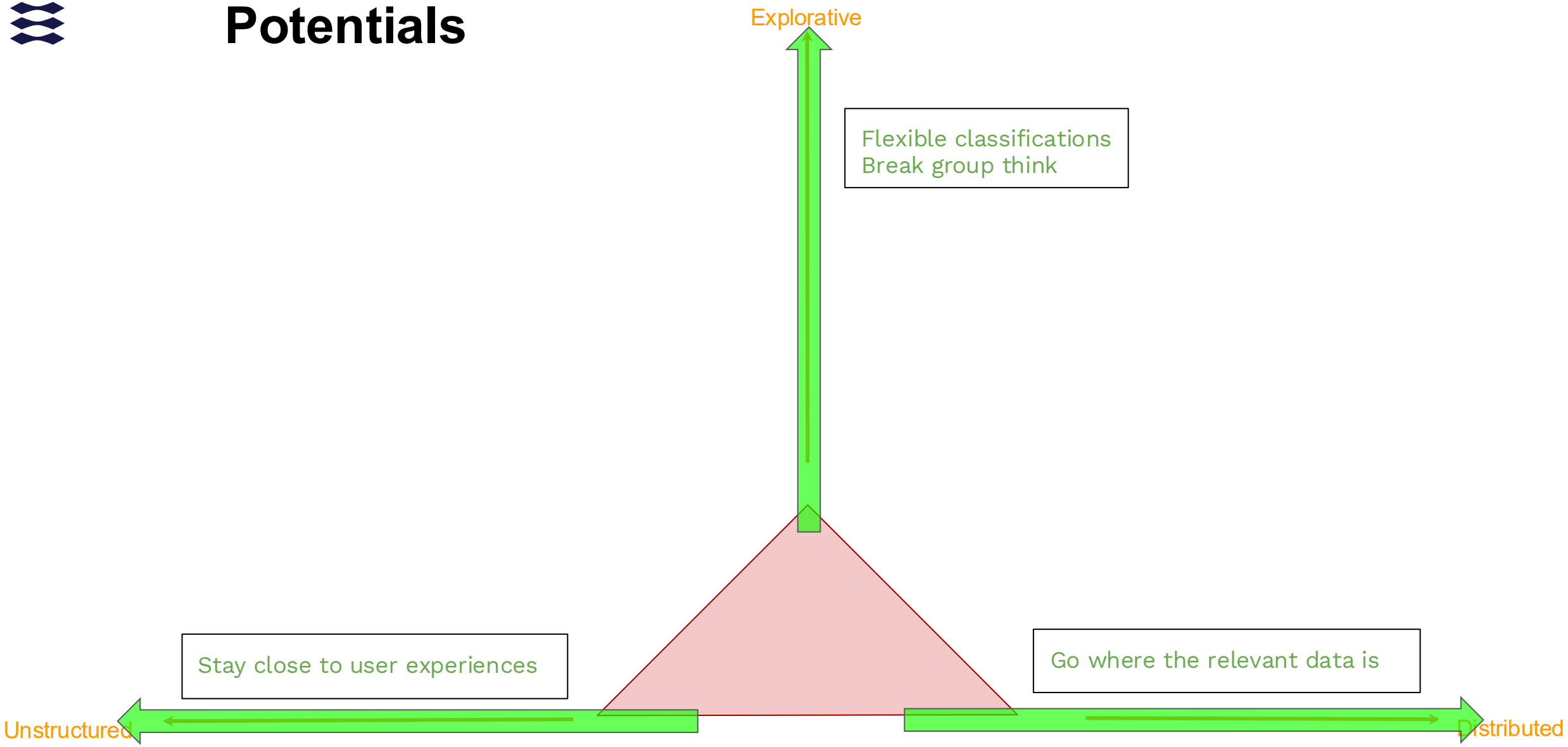
A new data-situation?



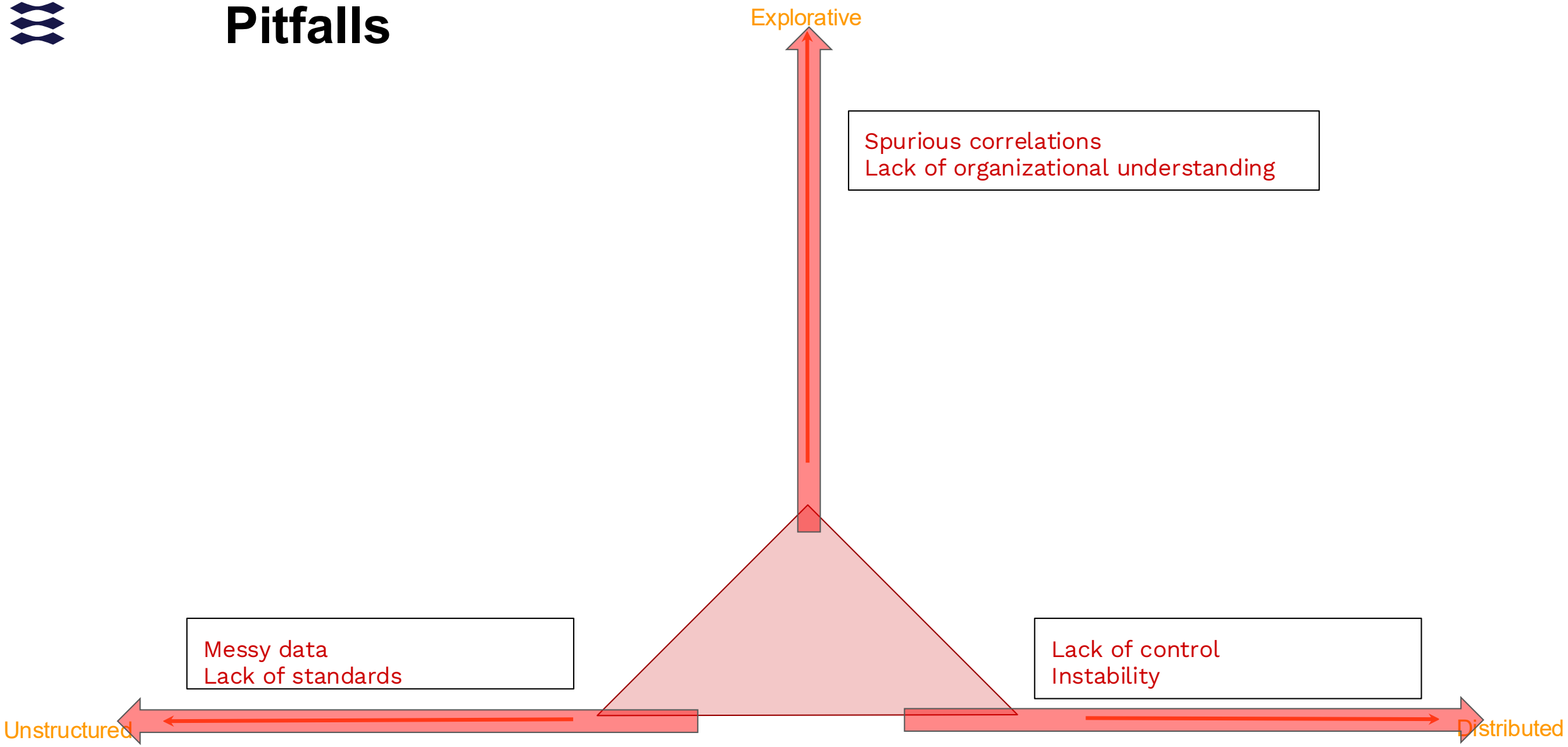
.... stimulated by AI



Potentials

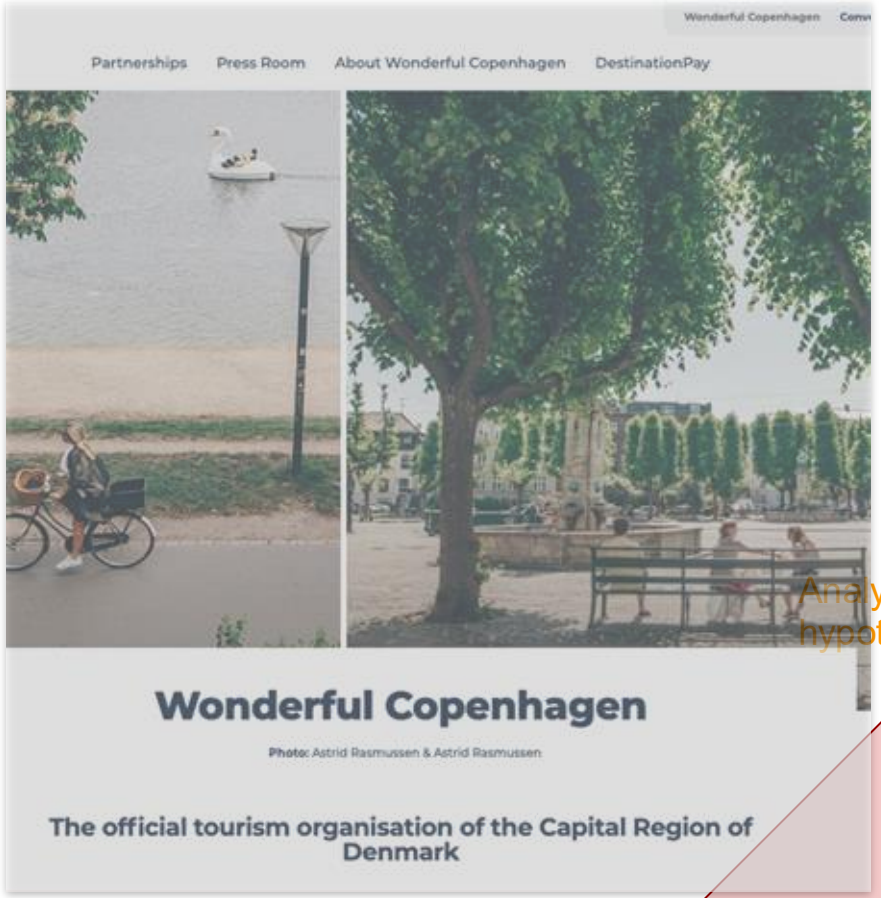

Explorative
Flexible classifications
Break group think
Stay close to user experiences
Go where the relevant data is
Unstructured
Distributed

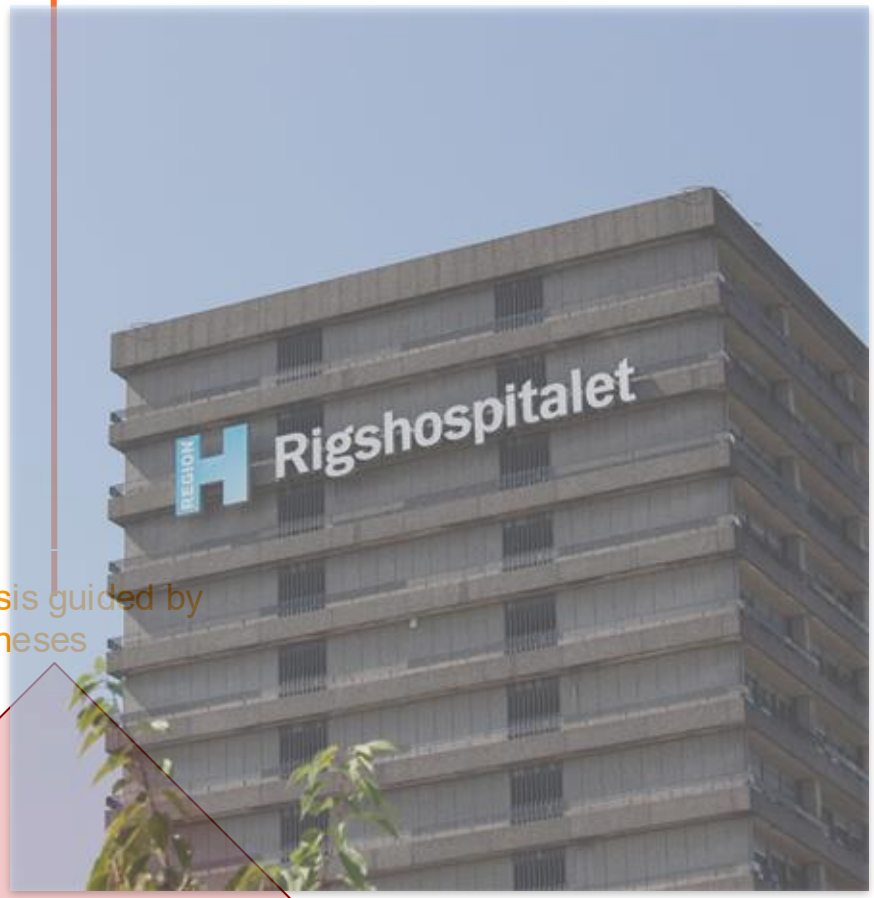
Pitfalls


Explorative
Spurious correlations
Lack of organizational understanding
Messy data
Lack of standards
Lack of control
Instability
Unstructured
Distributed

Relevance for cases?

Explorative





Analysis guided by hypotheses

Conventional structured data

Controlled and standardized data

Unstructured

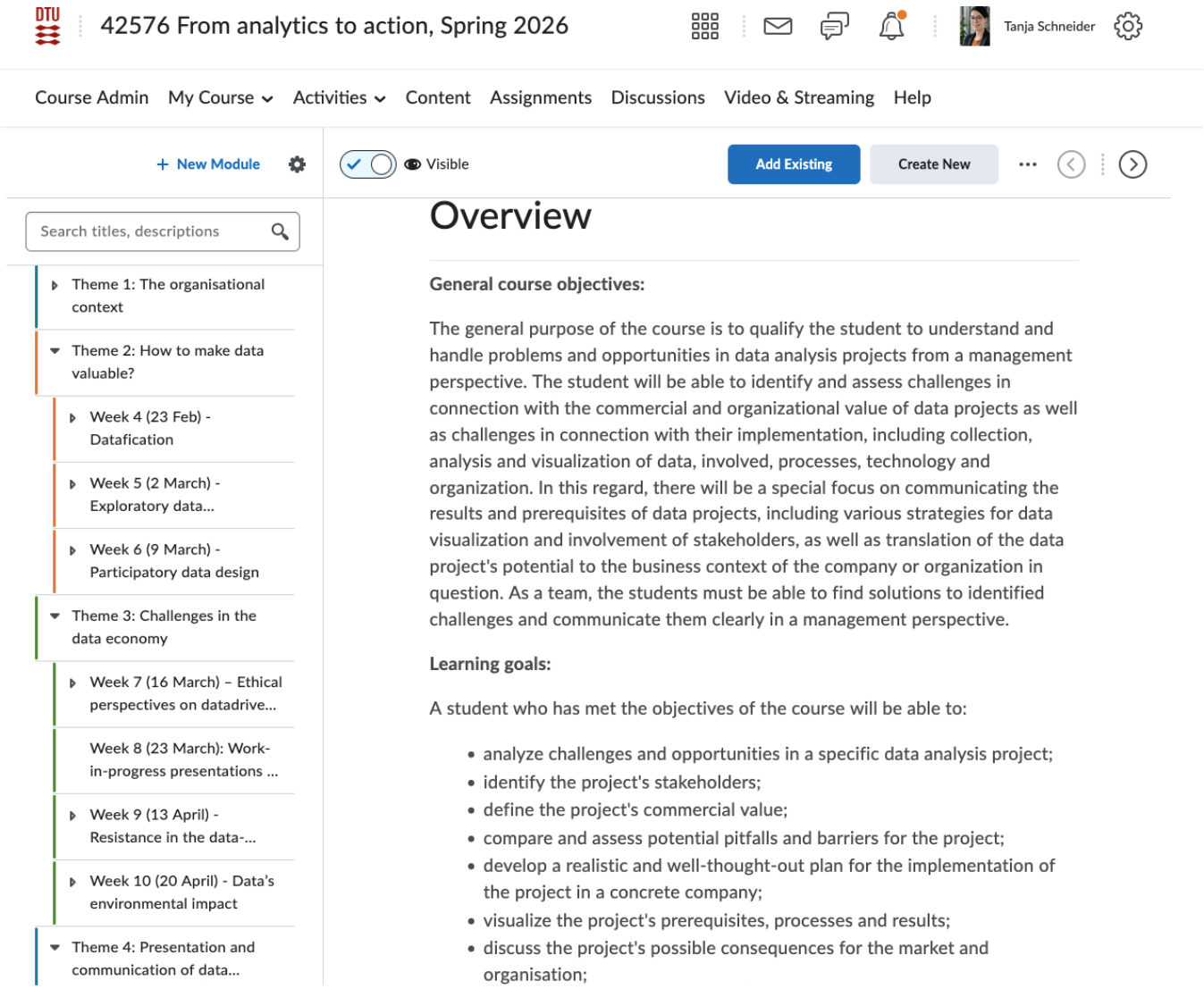
Distributed

The syllabus

- **4 Themes:**
  1. The organizational context
  2. How to make data valuable?
  3. Challenges in the data economy
  4. Presentation and communication of data projects

- Several guest lectures by researchers

- Cases/organisations



Typical session structure and overall workload

- **Plan of the day:**

  08:30 - 9:30 - Group check in and preparation for reading seminar.

  09:30-11:00 - Lecture, including exercises, guest lectures.

  11:00-12:00 - Reading seminar (facilitated by TAs)

- But be aware of different session formats on some days (check Learn)!

- **Workload:**
  - 5 ECTS course equals 140 hours of work.
  - We have class for 4 hours per week over 13 weeks, which equals 52 hours in class learning. So, there is 88 hours remaining for outside class course work.

Team work

- You work in teams of six students.
- **Teams will be formed in week 4 of the semester after case companies presented**

The reading seminars

We ask you to discuss the assigned readings in more detail with the TAs during lecture time (11:00 and 12:00).

The discussions will be facilitated by the Teaching Assistants.

Discussions centre on the general questions below and additional text specific questions per reading each week (see Learn):

- What is the most important issue in the readings as you see it?
- What did you find most interesting or unexpected?
- What is one core issue that the reading sheds light on that you would like to discuss in the seminar and why?

These **discussions will support you in your organisational anlysis of your cases and prepare you for the conceptual part of the oral examination.** We advise you to develop a glossary of the most important analytical concepts identified in the readings.

Rooms for Reading Seminars: Building 324, Rooms: 020 (36), 040 (60), and 50 (36)

Group work (1/2)

Cases and work-in-progress presentations:

- Starting from week 4: During and beyond the course time you will work in teams (max. 6 students) on an assigned case.
- The case will present the organisational context and their organisational challenge in class. They will also provide you with a dataset to address this challenge.
- We **ask you to conduct an exploratory instead of hypothesis driven data analysis! The organisation does not yet have the right framing of the question. This is your task**.
- You will be asked to **present an explanatory data visualisation that includes insights about the sector-specific issues and present at work in progress presentation**.

Group work (2/2)

- For this you must draw on your existing data analytics skill set (based on prior knowledge, see course prerequisite) and the **exploratory data analysis** and visualisation skills taught by Anders Koed Madsen.

- You will present your question and preliminary responses to the case organisations. In the **work in progress presentations** you will receive feedback from representatives of the organisation, lecturers, and TAs (no grading).

- For the **oral examination** you will further develop the presentation based on the feedback, will present an **explanatory data visualization**and will incorporate analytical concepts from the course readings. You will also be asked questions about the course content (syllabus) individually.

- Presence in the team presentations is required to pass the course.

The oral examination

- Oral examination: Oral exam with pitch of data visualization:
  - During the semester, a small pilot project is prepared in teams of 6 students, which is presented as an explanatory data visualization.
  - The visualization is handed in at the end of the semester (before the end of the 13 weeks) and brought to the exam, where the pilot project is pitched. Each group member presents their contributions.
  - All group members are then examined individually on the syllabus based on the questions the pitch gives rise to.
  - The explanatory data visualization is accounts for 25%, the pitch for 25%, and the subsequent oral examination for 50% in the overall assessment.

- Exam duration: Oral exam for 45 minutes (6 students), including pitch for a maximum of 12 minutes. The data visualization included in the pitch is submitted in advance with a maximum of 450 words.

- Evaluation: 7 step scale, external examiner.

- Assessment criteria: Consideration of organizational context, clarity of data analysis and visualization, quality of pitch, use and application of analytical concepts from course, critical reflection on the role of data analytics in organizational context.

Important dates

Date/s	Event
16 February	2 organizations will present their organisation, their challenge and available data to you
23 March	Student work-in-progress presentations to organisations (all team members need to be present)
11 May	Exam hand-in, 8:00 (morning)
13 and 18 May	Oral exams (detailed schedule to follow)

DTU